Nandheeswaran M

AWS Cloud & DevOps Engineer

Bangalore, India
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LinkedIn — Portfolio

Professional Summary

AWS Certified Solutions Architect – Associate (97%) with hands-on experience in designing, automating, and managing scalable cloud infrastructure on AWS. Strong expertise in Infrastructure as Code (Terraform), CI/CD engineering, Docker, Kubernetes (EKS), Linux system administration, and monitoring. Proven ability to implement DevSecOps practices, reduce infrastructure cost, and ensure high availability in production environments.

Technical Expertise

Cloud:	EC2, VPC, IAM, ALB, NLB, Auto Scaling, ECS, EKS, Lambda, S3, RDS, Aurora, Route53, CloudFront, CloudWatch
Infrastructure as Code:	Terraform (Modules, Workspaces, Remote Backend, State Locking), Ansible
CI/CD & DevOps:	Jenkins, GitHub Actions, Bitbucket, Maven, SonarQube, Nexus, Git branching strategies
Containers:	Docker, Kubernetes (Deployments, Services, Ingress, HPA), Helm, AWS EKS
Monitoring:	Prometheus, Grafana, CloudWatch, Log Analysis
Scripting:	Bash, Shell scripting, Python (Basic)
Operating Systems:	Linux (RHEL, CentOS, Ubuntu)

Professional Experience

System Engineer **Oct 2024 – Present**
Tata Consultancy Services (TCS)
Bangalore, Karnataka

- Managed and supported 20+ Linux production servers maintaining 99.9% uptime.
- Designed secure AWS network architecture including VPC, public/private subnets, NAT Gateway, Internet Gateway, and route tables.
- Implemented Auto Scaling Groups and Application Load Balancers for high availability deployments.
- Provisioned AWS infrastructure using modular Terraform with S3 backend and DynamoDB state locking.
- Built CI/CD pipelines integrating Git, Maven, SonarQube, Nexus, Docker, and Jenkins.
- Automated build, artifact management, and container deployment workflows.
- Deployed containerized applications to Kubernetes and AWS EKS clusters.
- Configured rolling updates, readiness/liveness probes, Ingress, and HPA.
- Implemented IAM policies and RBAC improving cloud security posture.
- Automated server patching and configuration management using Ansible playbooks.
- Reduced infrastructure cost by 15% through EC2 right-sizing and monitoring optimization.
- Implemented centralized monitoring dashboards using Prometheus and Grafana.
- Performed root cause analysis (RCA) for production incidents and resolved high-severity issues.
- Collaborated with development teams to streamline deployment workflows and reduce release time.
- Implemented environment-based deployment strategies (Dev, Staging, Production) using branching and infrastructure isolation.
- Configured secure IAM role assumptions and least-privilege access policies across AWS services.
- Integrated application logging and centralized log analysis to improve incident resolution time.
- Optimized Kubernetes resource requests and limits to enhance cluster performance and cost efficiency.

Key Projects

Terraform-Based AWS Infrastructure Automation Portfolio Link

- Developed reusable Terraform modules to provision VPC, EC2, IAM roles, and Security Groups.
- Implemented multi-environment deployment strategy (Dev, Staging, Production).
- Configured secure state management using S3 backend with DynamoDB locking.
- Applied Infrastructure as Code best practices including variables, outputs, and lifecycle rules.
- Enabled scalable architecture supporting Auto Scaling and load balancing.

End-to-End DevSecOps CI/CD Pipeline

- Designed automated pipeline integrating Bitbucket, Maven, SonarQube, Nexus, Docker, and Ansible.
- Implemented static code analysis and quality gates before production deployment.
- Secured credentials using Jenkins credential store and Ansible Vault.
- Reduced deployment time by 30% through automation.
- Implemented rollback mechanisms and version-controlled releases.

Kubernetes Deployment on AWS EKS

- Deployed containerized microservices to AWS EKS cluster.
- Configured Ingress controller, Services, HPA, and zero-downtime deployments.
- Implemented namespace isolation and RBAC.
- Monitored cluster metrics and application performance using Prometheus and Grafana.
- Optimized resource allocation for pods to improve performance.

DevOps Workflow on AWS EC2

- Built automated CI/CD pipeline for web applications hosted on AWS EC2.
- Integrated Jenkins, Git, Docker, and Ansible for build and deployment automation.
- Containerized applications to ensure consistency across environments.
- Automated infrastructure provisioning and configuration management.
- Implemented monitoring and logging for application health tracking.

Certifications

- AWS Certified Solutions Architect – Associate (Score: 97%)
- Professional Training in Linux, Git, Ansible, Jenkins, Docker, Kubernetes
- Actively preparing for AWS Developer Associate Certification

Education

Bachelor of Engineering — Electronics and Communication Engineering 2020 – 2024
Dr. Mahalingam College of Engineering and Technology, Anna University
CGPA: 8.9

SSLC — Maths & Biology
SRV Boys Higher Secondary School
Percentage: 84%

Awards & Recognition

- On-the-Spot Award at TCS for technical excellence and production support.
- Learning Achievement Award for academic excellence.